Skilling Experiment 2: Titanic Preprocessing Pipeline

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Machine Learning (25SC2107E)

MCQs — Libraries

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How 8 columns become 11

Group	Column	Columns out	Why
Numeric	`age`, `fare`, `sibsp`, `parch`, `is_alone`	5	already numbers
Categorical	`sex`	**2**	`sex_female`, `sex_male`
Categorical	`embarked`	**3**	`embarked_C`, `_Q`, `_S`
Ordinal	`pclass`	1	becomes 0, 1 or 2
Total		**11**	from 8

Key Insight

Only the **word** columns grow. `sex` (2 values) and `embarked` (3 values) become $2 + 3 = 5$ columns in place of 2, giving **3 extra**. And $8 + 3 = 11$.

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(a) `age` (b) `sex` (c) `fare` (d) `pclass`

MCQs — Counting the columns

36. `pclass` goes through `OrdinalEncoder`. How many columns does it produce?
(a) 1 (b) 2 (c) 3 (d) 0

Answer: (a)

37. If `embarked` had 5 ports instead of 3, the final count would be
(a) 11 (b) 13 (c) 9 (d) unchanged

Answer: (b)

38. Why does one-hot encoding use a separate column per value rather than the numbers 0, 1, 2?
(a) it is faster (b) so that no value is treated as bigger than another (c) to save memory (d) it is required by pandas

Answer: (b)

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Answer: (a), (c), (d)

MCQs — Fare before and after scaling

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